

# Guilherme Salles

US Citizen | Brazilian Citizen | EU Citizen

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## EDUCATION

### Carnegie Mellon University

Bachelor of Mathematical Sciences, Major in Computer Science

GPA: 3.88/4.0

Selected Coursework: Introduction to Mathematical Finance, Multidimensional Calculus, Principles of Imperative Computation, Mathematical Concepts and Proofs, Matrices and Linear Transformations, Fundamentals of Programming and Computer Science

Activities: Carnegie Mellon Informatics and Mathematics Competition (CMIMC), Quant Club, Putnam

### American Heritage Schools – Broward Campus

Diploma

Completed Coursework: Machine Learning & AI, Differential Equations, Multivariable Calculus, Linear Algebra,

AP Calculus AB-BC Competition, AP Statistics Competition

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## SKILLS

**Programming Languages and Tools:** Python, C, LaTeX, Linux, Github, Microsoft Azure, NumPy, Pandas, Scikit-learn, TensorFlow

**Languages:** English, Portuguese

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## EXPERIENCE

### SURA Researcher

Carnegie Mellon University

May 2026 — July 2026

- Conducted research on the Theory of Computation using Raymond Smullyan's book *To Mock a Mockingbird* under the guidance of Professor Richard Statman.
- Learned and experimented with songbirds, a theoretical language used by Smullyan to explore the theory of computation.
- Created intermediary proofs to show that we cannot find a fixed-point combinator (Sage bird) using songbirds  $B$ ,  $I$ , and  $M$ , establishing the inability to construct songbirds  $T$  and  $K$  in a restricted environment.

### AI Software Engineer Intern

Starwood Property Trust

Miami Beach, FL

May 2025 — June 2025

- Using Python and the OpenAI library, trained and evaluated a custom GPT-based classifier on a large collection of PDF-extracted text to identify legal documents, achieving 100% accuracy on an internal test set.

### Teaching Assistant and Test Writer

Mu Alpha Theta Honor Society, American Heritage Schools

Plantation, FL

August 2022 — July 2025

- Organized 50+ classes, 15+ workshops, and wrote 15+ math competition tests in Geometry, Algebra 2, Precalculus, Calculus, and Linear Algebra.
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## PROJECT EXPERIENCE

**Brief.ly** – Application, Hackathon Submission

NOVA Hackathon, Scotty Labs

Pittsburgh, PA

November 2025

- Developed a GPT-based summarization module to convert scraped news articles into structured summaries for personalized newsletters in Python.
- Integrated a multi-step pipeline spanning web scraping, summarization, and automated email delivery.

**Tetris AI** – AI Model

American Heritage Schools - Broward Campus

Plantation, FL

December 2024

- Designed and trained a Tetris agent in Python using a genetic algorithm, clearing 100,000+ lines in a single run.
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## HONORS & AWARDS

**CMIMC Integration Bee 1<sup>st</sup> Place**

March 2025

**44 "Top 5" Awards in Mu Alpha Theta Competitions**

January 2022 — July 2025

**Three-time AIME Qualifier (Best Score: 7)**

October 2022 — February 2025